

Geopolitical Risk Scoring and Anomaly Detection

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Introduction

- Geopolitical risk is the threat or escalation of wars, terrorism, and tensions between states, which can disrupt international relations.
- The leading measure for Geopolitical risk is the GPR index.
- Limited coverage: GPR scores only 44 countries.
- News-only: GPR relies only on news coverage, it ignores a country's fundamentals.

This presents an opportunity: score geopolitical risk for every country, based on both news and fundamentals, and monitor it to catch crises early

Methodology : Pipeline

Data collection	• Collected 9 datasets, (2015–present) • Collection conducted via APIs/BigQuery/downloads/email request to the providers.
Data preparation	• Cleaned each source, selected and engineered features, standardized to a common country-year grid and merge all 9 into one master table.
Modeling	• Model 1 (<i>supervised</i>) : trained on the 68 features to predict the GPR index, compared XGBoost, Random Forest, and ElasticNet. • Model 2 (<i>unsupervised</i>) : trained on news signals to learn each country's baseline and flag anomalies, compared z-score, Isolation Forest, and LOF.
Validation	• Model 1: hold out whole countries (GroupKFold) and whole years (expanding window) retrain on the rest, predict the held-out. • Model 2: check flags against a list of known crisis onsets.
Output	• Model 1: Geopolitical risk scores for all 247 countries. • Model 2: monthly unrest alerts per country.

Data

Source	What it measures	Example features	Features
GPR	Geopolitical risk level : Target	GPRC	1
WDI	Economy	GDP per capita, inflation, military spending	15
V-Dem	Governance and democracy	regime type, level of corruption,	5
GSDB	Sanctions	active sanctions cases	9
UN GA Voting	Diplomacy	agreement with US / Russia	4
UNHCR	Forced displacement	Number of refugees	4
UN Comtrade	Trading	total trade, export concentration (HHI)	20
SIPRI	Arms transfers	arms imports / exports number of suppliers	3
GDELT	News events	arms imports / exports (TIV), number of suppliers	8

Model 1 : Risk Scorer (Supervised)

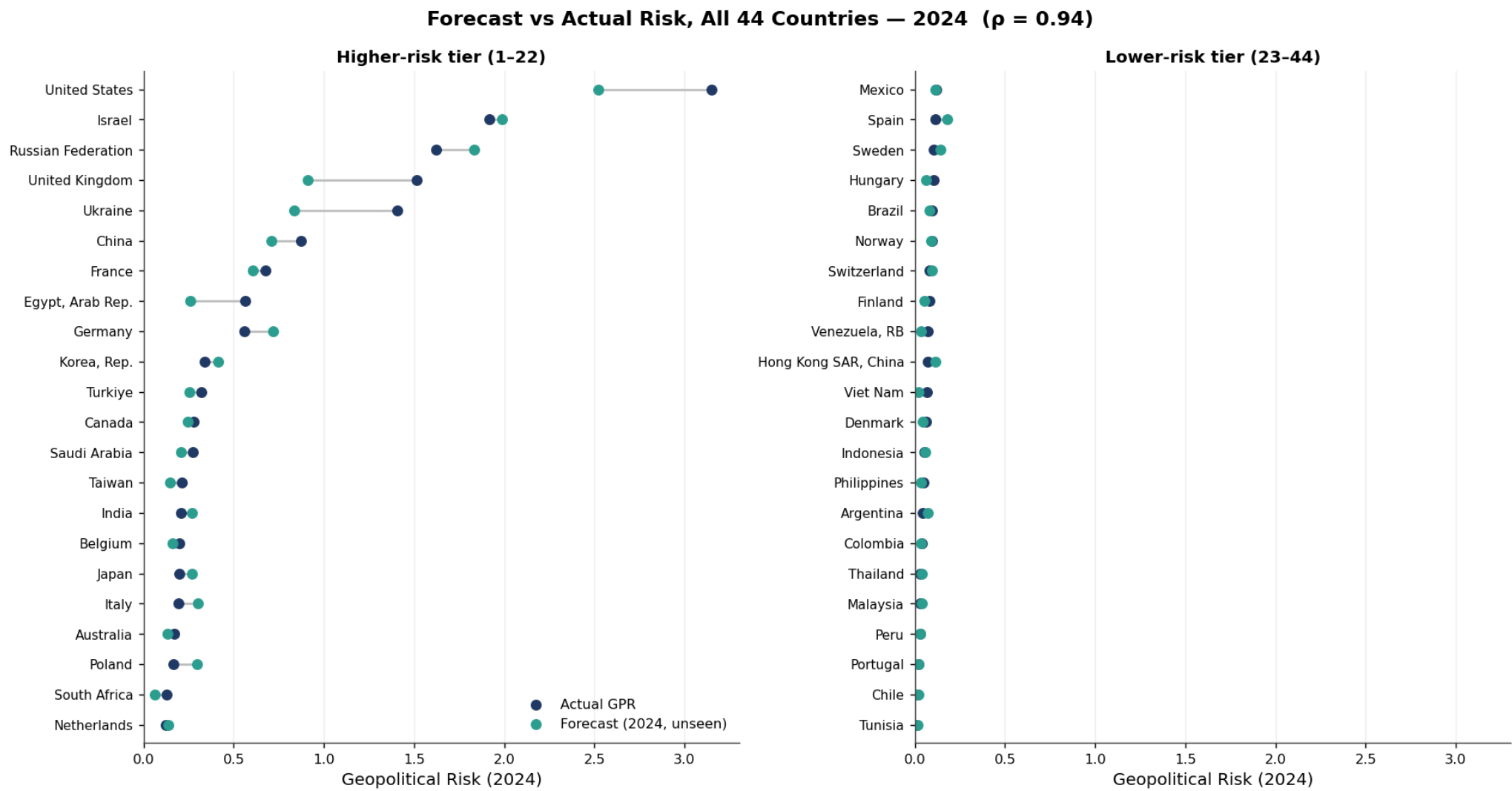
Approach

- Goal : Predict the GPR index for every country.
- Target : The GPR index, log-transformed (it's skewed).
- Data : Trained on the 44 labeled countries × 10 years (2015–2024) = 440 rows, with 68 features.
- Models : XGBoost regression vs Random Forest vs ElasticNet.

Validation

Model	Spearman (unseen countries)	Spearman (unseen years)	MAE (unseen countries)	MAE (unseen years)
XGBoost	0.811	0.911	0.16	0.11
RandomForest	0.743	0.906	0.178	0.119
ElasticNet	0.639	0.845	0.241	0.139
Baseline (predict mean)	-	-	0.274	0.275

Model 1 : Risk Scorer



Note. 2024 is forecast by a model trained only on 2015–2023. Both panels share one x-scale; lower-risk countries cluster near zero because their risk — and their forecast error — are small.

Model 2 : Anomaly Detector (Unsupervised)

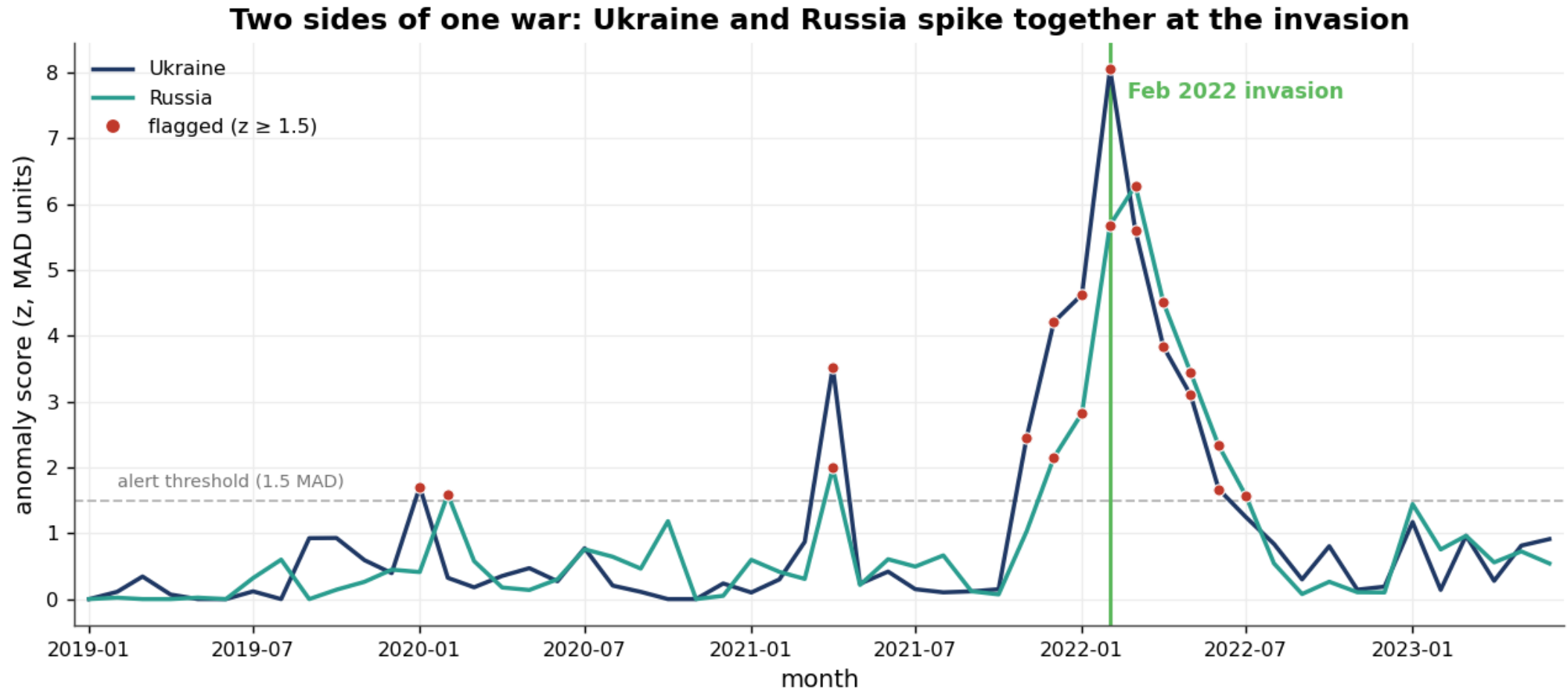
Approach

- Goal: Flag sudden unrest for countries.
- Target: None. The model would detect deviations from each country's own baseline.
- Data: monthly GDELT, all countries, 32k country-months, 7 signals.
- Models: z-score vs Isolation Forest vs LOF.

Validation

Method	Events flagged	Baseline (Random)	P-value
Z-score	43/50	20	<0.001
Isolation Forest	43/50	21	<0.001
LOF	17/50	15	0.34

Model 2 : Anomaly Detector



Conclusion and Future Works

Project contributions

- Risk scores for all 247 countries
- Early-warning monitor for unrest

Future works

- Improving data quality (duplicated news, misinformation)
- Improving Model 2 precision by reducing false alarms
- Deployment

References

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- Yonamine, J. E. (2013). *Predicting future levels of violence in Afghanistan districts using GDELT*.

Datasets

- GPR : Caldara, D., & Iacoviello, M. (2022). *Geopolitical Risk Index* (target).
- GDELT : The GDELT Project. *Global Database of Events, Language, and Tone* (Leetaru & Schrodt, 2013).
- World Bank WDI : World Bank. *World Development Indicators*.
- V-Dem : Coppedge, M., et al. (2026). *V-Dem Dataset v16*.
- SIPRI : *SIPRI Arms Transfers Database*.
- UN Comtrade : United Nations. *UN Comtrade Database*.
- GSDB : Felbermayr et al. (2020); Yalcin et al. (2025). *Global Sanctions Data Base (R4)*.
- UNHCR : *UNHCR Refugee Population Statistics Database*.
- UN GA Voting : Bailey, Strezhnev & Voeten (2017). *UN General Assembly Voting Data*.

Thank you for your attention